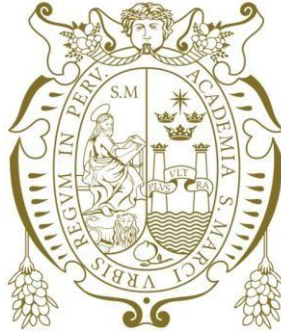


UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Doctoral Program in Deep Tech, Artificial



Intelligence & Emerging Technologies

**Research Methods and Scientific Integrity in AI and
Advanced Technologies**

PARADIGM JUSTIFICATION STATEMENT DELIVERABLE No. 1

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Statement of Epistemological Paradigm Justification

“A Verifiable Generative Artificial Intelligence Model for Contractual Clause Recommendation in Regulated Fiduciary Institutions”

Researcher: Jesús Vicente Inca Cárdenas — 2026

1. Research paradigm and its ontological grounding

This research is framed within the Design Science Research (DSR) paradigm, philosophically grounded in critical realism. This choice responds to the nature of the object of study: the construction and evaluation of a novel artefact a verifiable generative artificial intelligence model for contractual clause recommendation whose knowledge is produced not merely by observing a pre-existing reality, but by rigorously designing, building, and evaluating a solution to a relevant, unresolved problem in the regulated fiduciary domain. Unlike purely descriptive or explanatory paradigms, DSR generates knowledge through the artefact itself and the design principles derived from it (Hevner, March, Park & Ram, 2004; Peffers, Tuunanen, Rothenberger & Chatterjee, 2007; Gregor & Hevner, 2013).

Ontological dimension. A critical realist stance (Bhaskar, 2008 [1975]) is adopted, distinguishing the domain of the *real* the generative mechanisms underlying socio-technical systems, such as SBS regulatory structures, the legal logic of the clauses, and the behaviour of generative models from the domain of the *actual* (the events that effectively occur, such as clause generation or the emergence of a hallucination) and the domain of the *empirical* (what the researcher is able to observe and measure). This stratification is well suited to a complex phenomenon in which technical mechanisms (augmented retrieval, verifiability, and hallucination control) coexist with institutional ones (traceability and regulatory auditability requirements). The reality studied exists objectively and independently of the researcher, yet remains only partially and fallibly knowable.

2. Why critical realism rather than pragmatism

Pragmatism is the more common philosophical underlabourer for Design Science Research, so the choice of critical realism warrants an explicit defence. Three considerations justify it, and each strengthens rather than complicates the argument.

First, the object of study is genuinely stratified. The generative mechanisms the SBS regulatory framework, the legal logic of the clauses, and the statistical behaviour of the model are real and causally efficacious even when they are not actualised in a particular event or observed by the researcher. Critical realism supplies exactly this real/actual/empirical stratification; a strictly pragmatist ontology, oriented to consequences and workability, does not distinguish these strata and therefore cannot account for mechanisms that operate whether or not they are exercised.

Second, verifiability presupposes a mind-independent foundation. For an output to be “traceable to a contrastable foundation,” that foundation the norm, the precedent, the source clause must exist and possess its content independently of the beliefs of the model, the user, or the researcher. A pragmatist reading would tend to collapse verifiability into usefulness or user acceptance, which is precisely what a

regulated fiduciary domain cannot permit: a clause is not well founded because it is found convincing, but because it corresponds to an identifiable and contrastable source.

Third, fallibilism fits a stochastic artefact evaluated against a contestable standard. Critical realism’s “modified objectivity” lets me claim knowledge of these mechanisms while acknowledging that such knowledge is partial and revisable the appropriate posture when the artefact is probabilistic and the legal gold standard is itself open to reasoned disagreement. This said, the pragmatic maxim central to DSR is retained at the methodological level: the utility of the artefact is a criterion of a good design. The philosophical difference is that “it works” is treated as evidence about the underlying mechanisms rather than as the ultimate arbiter of truth about them.

3. Epistemological dimension

Knowledge is constructed through an iterative process of design and evaluation, in which the validity of claims is established by the artefact’s demonstrable capacity to solve the problem and by the empirical evidence supporting its performance. The researcher assumes a modified objectivity: acknowledging that these constructions are perfectible approximations, bias is minimised through source triangulation, expert judgement, the use of an annotated gold standard, and controlled comparison of the artefact against baselines. Verifiability, the central axis of this thesis, is, epistemologically, the requirement that every system output be traceable to a contrastable foundation, thereby translating the principle of refutability proper to rigorous scientific thought into the technical plane.

4. Preliminary research question

The problem is both a design problem and an evaluation problem, so the research question is stated in two coordinated parts.

Main question. How can a verifiable generative AI artefact be designed to recommend contractual clauses in a regulated fiduciary institution so that every recommendation is traceable to a contrastable normative or contractual source and does such an artefact outperform a standard retrieval-augmented generation (RAG) baseline in verifiability and hallucination control without loss of recommendation quality?

RQ1 (design). What design principles govern a verifiable clause-recommendation artefact whose outputs are attributable to identified sources in an SBS-supervised fiduciary setting?

RQ2 (evaluation). Does the artefact achieve significantly higher attribution quality and significantly lower hallucination rates than (a) a no-retrieval baseline, (b) standard RAG, and (c) current institutional practice, while maintaining recommendation quality?

5. Implications for data, methods, contribution, and target venues

Data. An anonymised corpus of fiduciary contracts and clauses; a regulatory corpus (applicable SBS norms and circulars, Ley 26702, and related provisions); and an annotated gold standard linking recommendations to supporting sources (see §7). All contract material is de-identified before use (see §9).

Methods. The DSR cycle after Peffers et al. (2007) problem identification, objectives of the solution, design and development, demonstration, evaluation, and communication articulated with a predominantly quantitative evaluation based on controlled comparison against several baselines and on human annotation.

Contribution. The contribution is dual, in the sense of Gregor & Hevner (2013): a prescriptive contribution nascent design principles for building verifiable generative systems in regulated domains and an empirical contribution evidence of the effect of an explicit verifiability layer on attribution quality and hallucination control.

Target venues. Information-systems outlets receptive to DSR (e.g., MIS Quarterly, Journal of the AIS, European Journal of Information Systems, and the DESRIST conference); AI-and-law outlets (e.g., the journal Artificial Intelligence and Law, and the ICAIL/JURIX conferences); and NLP venues with tracks on attribution and factuality. Final venue selection will follow the maturity of the results.

6. Operationalising verifiability

Verifiability is the central construct and must be measured rather than asserted. It is operationalised through the notion of attribution and made concrete with established metrics.

Conceptual definition. An output is treated as verifiable when it is Attributable to Identified Sources in the sense of Rashkin et al. (2023): a proposition s is attributable to a source P when a reasonable reader would affirm “According to P , s .” This test furnishes a replicable, source-anchored definition of what it means for a recommendation to rest on a contrastable foundation.

Metrics. Attribution is quantified with citation precision and citation recall in the sense of Gao et al. (2023): recall captures whether the cited sources jointly support the generated clause or assertion, and precision captures whether each cited source is genuinely relevant (penalising irrelevant citations). These are complemented by span-level attributability judged by human raters, who decide, for each asserted proposition, whether the cited span actually supports it.

Operational definition of hallucination. In this domain, a generated proposition counts as a hallucination when it is (i) not attributable to any provided source (unsupported), (ii) attributed to a source that does not in fact support it (misattributed), or (iii) in contradiction with the governing regulatory framework. Following the intrinsic/extrinsic distinction of Ji et al. (2023), the hallucination rate is reported as the proportion of generated propositions that fail attribution, disaggregated by type.

7. The gold standard

Because legal correctness is contestable, the gold standard cannot rest on a single author. It will be constructed by at least two preferably three qualified annotators, namely lawyers with demonstrable experience in fiduciary matters and financial regulation, working from a written annotation guideline. Inter-annotator agreement will be reported with an appropriate statistic Cohen’s κ for two raters or Krippendorff’s α when three raters or partial coverage are involved with a minimum acceptable threshold stated in advance (for example, $\alpha \geq 0.67$, with ≥ 0.80 preferred). Disagreements will be resolved through a documented adjudication procedure, whether by consensus or by a senior third expert. The gold

standard is understood to capture defensible, consensus positions rather than a single incontrovertible ground truth, and this limitation is stated explicitly.

8. Baselines and statistical analysis

Standard RAG is the right primary comparator, but it is not sufficient on its own to isolate the contribution of the verifiability layer, nor to sustain the operational-impact claim. The evaluation therefore contrasts four conditions:

B0 — No-retrieval (closed-book) baseline: isolates the contribution of retrieval itself.

B1 — Standard RAG: the primary comparator.

B2 — Proposed artefact: RAG augmented with the verifiability/attribution layer and hallucination control.

B3 — Current institutional practice: a human baseline, required for the operational-impact claim.

Statistical design. The unit of analysis is the individual clause-recommendation instance; the sample size is fixed by a prior power analysis targeting a meaningful difference in attribution and hallucination proportions. Because generative outputs are stochastic, each item is generated k times per system (with fixed seeds where feasible), and results are reported as means with variance across repeated runs rather than as a single pass. Systems are compared on the same items using paired procedures suited to proportions (for example, McNemar’s test or a paired bootstrap), with item-level random effects considered to account for item variance. Family-wise error across the set of metric $\times$ baseline contrasts is controlled (Holm–Bonferroni) or the false-discovery rate is bounded (Benjamini–Hochberg). Effect sizes and confidence intervals are reported alongside p -values.

9. Confidentiality, legal risk, and human oversight

Legal exposure. The artefact operates on fiduciary contracts within an SBS-supervised institution, which engages two distinct regimes: banking secrecy under Ley 26702 (Ley General del Sistema Financiero y del Sistema de Seguros, art. 140) and personal-data protection under Ley 29733 (Ley de Protección de Datos Personales) and its regulation. Clause text may embed protected client information, so any processing must be lawful under both regimes.

Consequences for design. Sending clause text to an external model API constitutes disclosure to a third party and is generally not admissible without a robust legal basis. The honest response is therefore: (a) de-identification or pseudonymisation of all contract text used in development and evaluation; (b) a preference for local, on-premise deployment using a self-hosted model, so that no client data leaves the institution’s controlled environment; and (c) where any external processing is truly unavoidable, a data-processing agreement and a data-protection impact assessment before use.

Human authority. The artefact is explicitly advisory. A qualified legal officer retains authority over any clause that is ultimately adopted; the system supports professional judgement but does not replace it. This is stated as a binding design constraint and an ethical safeguard, and it also bounds the institution’s liability.

10. Transferability

The artefact is instantiated in a single SBS-supervised institution, yet the intended knowledge contribution is the set of design principles, not the particular instance. Transferability is therefore claimed at the level of abstraction — from instantiation to nascent design theory — consistent with Gregor & Hevner (2013). To warrant the claim rather than merely assert it, three mechanisms are foreseen: an explicit abstraction step from the instantiation to stated design principles; validation by an expert panel drawn from other supervised entities, who judge applicability to their own settings; and, ideally, a second-site demonstration or, minimally, an articulated account of the boundary conditions under which the principles hold. In the absence of a second site, transferability is presented as analytical generalisation, not statistical generalisation.

11. The novelty claim and its structured search

The assertion that neither the literature nor the Peruvian State offers a generative AI model guaranteeing the verifiability of contractual clauses in regulated fiduciary settings is a strong claim and will be supported by a structured, auditable search rather than left as a statement. This entails a systematic-review protocol (PRISMA-style): named databases (for example, Scopus, Web of Science, IEEE Xplore, the ACL Anthology, and SSRN), an explicit search string, stated inclusion and exclusion criteria, and a defined date range; complemented by a review of applicable SBS regulation and of Peruvian State initiatives in public-sector AI. Reporting the protocol makes the gap claim reproducible and defensible.

12. One doubt or tension

The most serious tension I currently hold open is that verifiability is not the same as correctness. My central construct measures whether an output is traceable to, and supported by, an identified source; but a clause can be perfectly attributable to a source that is itself outdated, inapplicable, or legally mistaken, while a legally sound clause might be penalised for weak surface attribution. Optimising attribution metrics could, at the margin, reward what is “citable” over what is “correct.” Critical realism sharpens rather than dissolves this tension: the real (the correct legal content) is not identical to the empirical (the measured attribution). Two commitments follow. First, correctness will be evaluated as a dimension distinct from attribution, not folded into it. Second, the thesis will state plainly that verifiability is a necessary but not sufficient condition for a trustworthy recommendation. I flag this as an open tension that the evaluation design must confront rather than conceal.

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